

Ming-Fen (Jing) Chung

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SUMMARY

VR/XR Developer focused on designing systems that translate physical signals into meaningful emotional experiences in immersive environments. Across four ETC projects, I have worked with breath, haptics, and real-world objects as interactive media. Background spans Unity/C#, computer vision, custom hardware pipelines, and firmware engineering.

WORK EXPERIENCE

Cooperative Tangible MR Escape Room Experience

Jan 2026 – Present

MR Developer & Programmer · Carnegie Mellon ETC

- Designed and built a computer vision pipeline from scratch (**OpenCV + ArUco**) to bind **50+** real-world household objects to in-game events in real time — no existing team knowledge of physical-virtual object tracking to build from
- Resolved spatial drift in multi-user sessions by integrating **Meta's MRUK** for scene understanding and anchoring, ensuring both players share a stable, consistent virtual space

In The End

Sept – Dec 2025

Technical Designer, Technical Artist & Programmer · Carnegie Mellon ETC

- Originated a novel interaction mechanic where players identify characters by decoding **5** distinct haptic heartbeat signatures, validating through playtesting whether tactile feedback could serve as the primary emotional driver
- Built the **Haptic Abstraction System** from first principles — abstracting device-specific logic so the team could begin testing before hardware arrived, then extended it so designers could swap haptic patterns without touching code
- Built a reusable shader graph library with **8+** unified-architecture visual effects, cutting art production time across the team

Xhaler

Jan – May 2025

Producer & System Architect Programmer · Carnegie Mellon ETC

- Worked with no reference implementation for breath-as-input: designed and integrated a **custom respiratory belt + Arduino pipeline** from hardware signal to gameplay response, making breath the game's sole control method
- Identified motion sickness as the primary experience risk early, redesigned locomotion from scratch using physics-based movement, and cut sickness reports by **60%**
- Optimized Quest 3 performance from unstable to sustained 60+ fps via efficient runtime scene loading/unloading; built a **reusable VR framework** adopted across 2+ subsequent ETC projects

Silicon Motion Technology Corp.

Zhubei, Taiwan

Principal Firmware Engineer

Jan 2020 – June 2024

- Architected** foundational codebase and **compiler infrastructure**, significantly reducing build cycle time
- Authored onboarding program that trained **10+** engineers on a new toolchain

FutureTown Inc.

Taoyuan, Taiwan

Gameplay Programmer

Oct 2018 – Aug 2019

- Diagnosed unnatural avatar motion artifacts across 2 shipped VR titles and reworked IK joint rotation constraints to fix them
- Designed a client-facing interactive VR car showcase with 5 distinct interaction modes

SELECTED PROJECTS

GO! KART VR

Aug 2019 – Dec 2020

Indie Game Designer & Programmer · Side Project

- Showcased at Korea VR Festival 2019 — a multiplayer VR racing game with motion-feedback chairs
- Implemented **Photon** networking with player sync, item events, and custom host migration

TECHNICAL SKILLS

Game Engines & Platforms: Unity, Unreal Engine 4/5, Meta Quest, HTC Vive

Languages: C#, C/C++, Python

Networking & Multiplayer: Photon (PUN), Netcode for GameObjects

XR & Haptics: OpenCV, ArUco, bHaptics, Unity Shader Graph

Hardware & Tools: Arduino, Git, Perforce, Maya, Blender

EDUCATION

Carnegie Mellon University

Master of Entertainment Technology

Pittsburgh, PA

Expected May 2026

National Cheng Kung University (NCKU)

M.S. & B.S. in Computer Science and Information Engineering

Tainan, Taiwan

2017 / 2015